

Probability and Probability Distributions

$6_{(3)}$

Tree diagrams

Tree diagrams

Tree diagrams show all the possible outcomes of an event and calculate their probabilities. All branches must add up to 1 when adding downwards. This is because the probability of something not happening is 1 minus the probability that it does happen. Multiply going across a tree diagram. Add going down a tree diagram.

Tree diagrams

Example 1:- There are 5 lemon and 4 strawberry sweets in a bag. Haile takes out a sweet at random, writes down its flavor and puts it back into the bag. Then Haile takes out a second sweet, at random, and writes down its flavor.

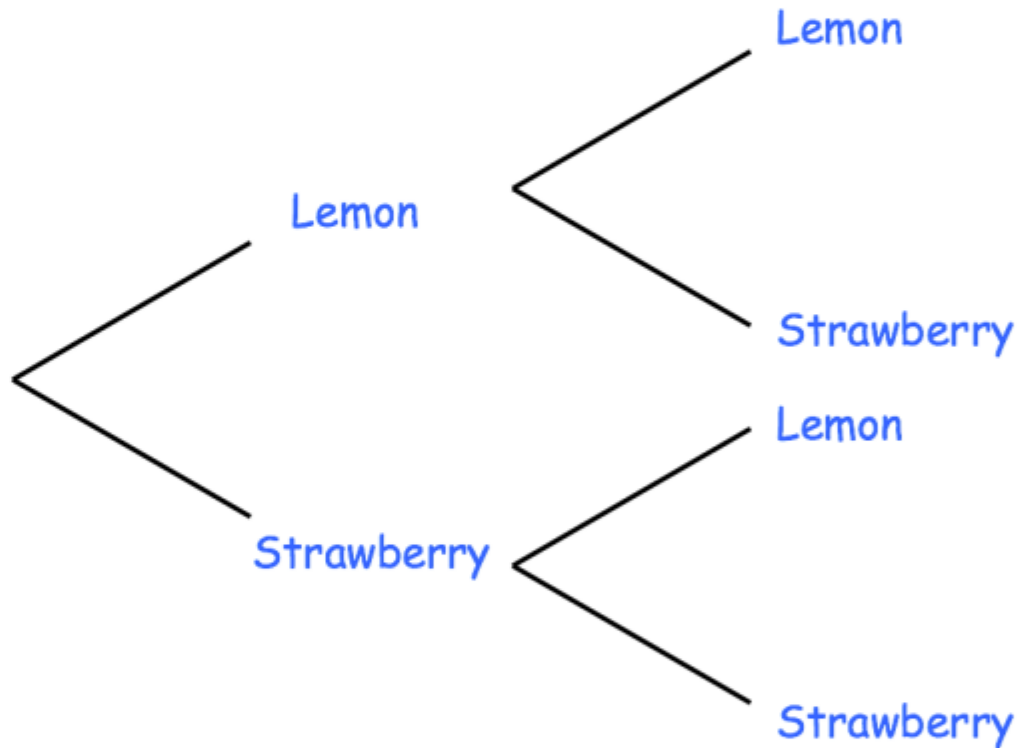
Tree diagrams

- (a) Construct the tree diagram
- (b) Find the probability of two lemon sweets
- (c) Find the probability of one of each flavor.
- (d) Find the probability of no lemon sweets

Tree diagrams

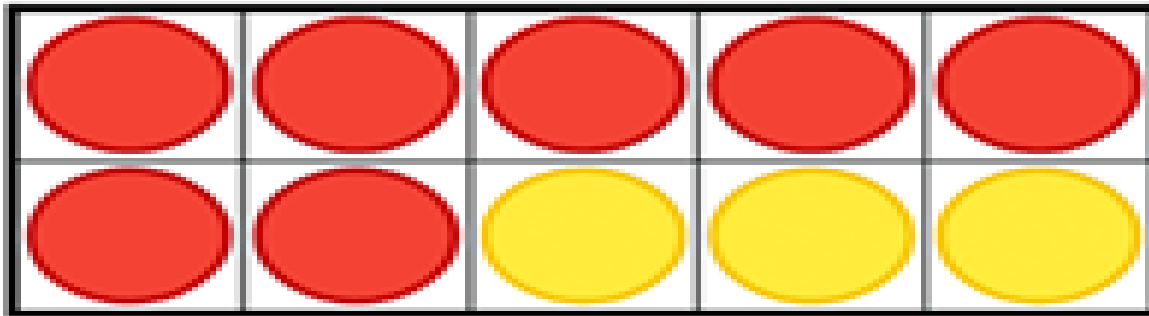
1st Sweet

2nd Sweet



Tree diagrams

Example 2:- There are 10 counters in a bag, 7 are red and the rest of yellow. Elias takes out a counter at random and records its color. Without replacement, Elias takes out another counter, at random.



Tree diagrams

- a) Complete the tree diagram
- b) Find the probability that both counters are different colors.
- c) Find the probability that both counters are the same color.

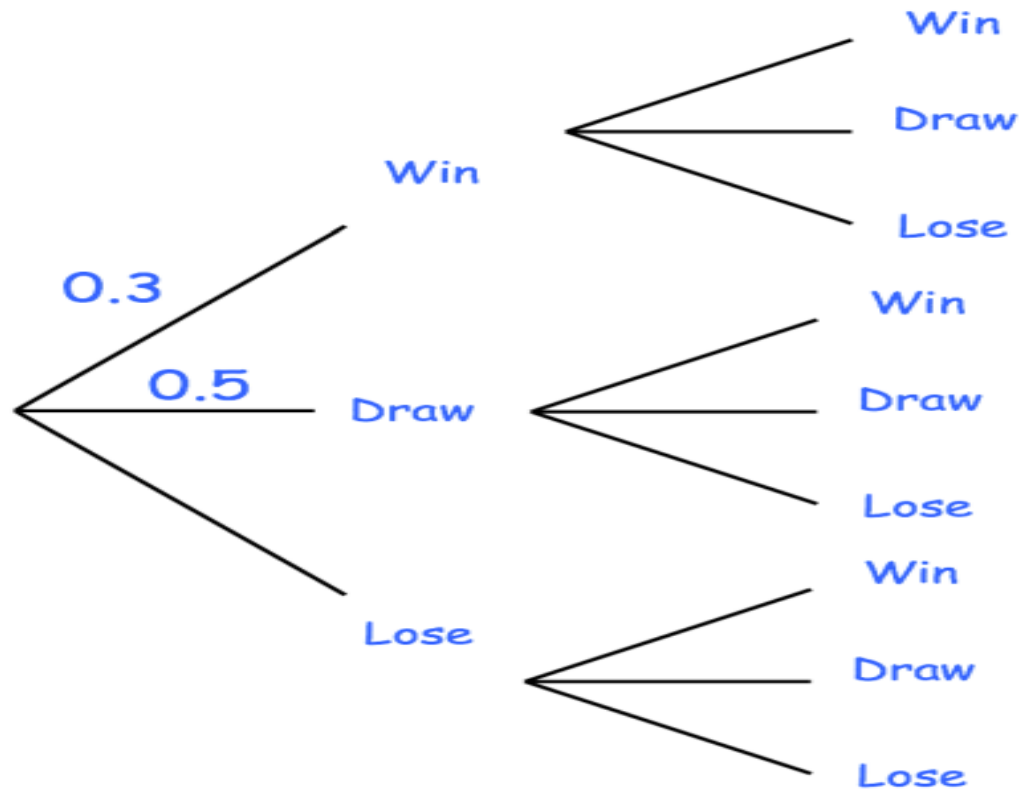
Tree diagrams

Example 3:- A football team has two remaining matches. The probability of winning a match is 0.3, while the probability of drawing is 0.5. A win earns the team 3 points, a draw earns 1 point, and a loss earns no points. Create a tree diagram to represent these outcomes and their probabilities.

Tree diagrams

first match

second match



Any Questions



Any Questions



Thank
you